

Assignment-1

1) Define Machine learning and write any 20 application of it.

⇒ Machine learning is subset of Artificial intelligence focused on the development of algorithms and models that enables computers to learn and make predictions without being explicitly programmed.

Applications :-

Manufacturing :-

- 1) predictive maintenance
- 2) warranty reserve estimation
- 3) Propensity to buy
- 4) Demand forecasting
- 5) Process optimization.

Retail :-

- 1) Predictive inventory planning
- 2) Recommendation engines
- 3) Upsell marketing
- 4) Market segmentation
- 5) Customer ROI

Healthcare & Life science :-

- 1) Diagnostic diagnostics from real-time patient data
- 2) Disease identification
- 3) Patient triage optimization
- 4) Proactive health management.
- 5) Healthcare providers sentiment analysis.

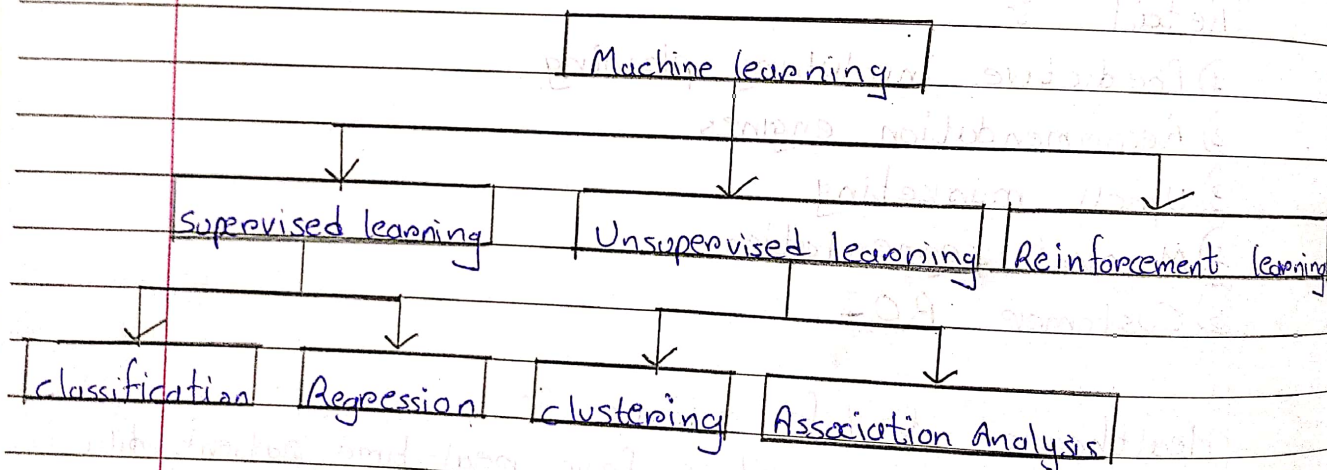
Financial services :-

- 1) Risk analytics
- 2) Customer segmentation
- 3) Cross-selling and up-selling
- 4) Sales and marketing
- 5) Credit worthiness evaluation

Travel and Hospitality :-

- 1) Aircraft scheduling
- 2) Dynamic pricing
- 3) Social media
- 4) Feedback analysis
- 5) Customer complaint resolution

2) Explain types of machine learning with proper diagram

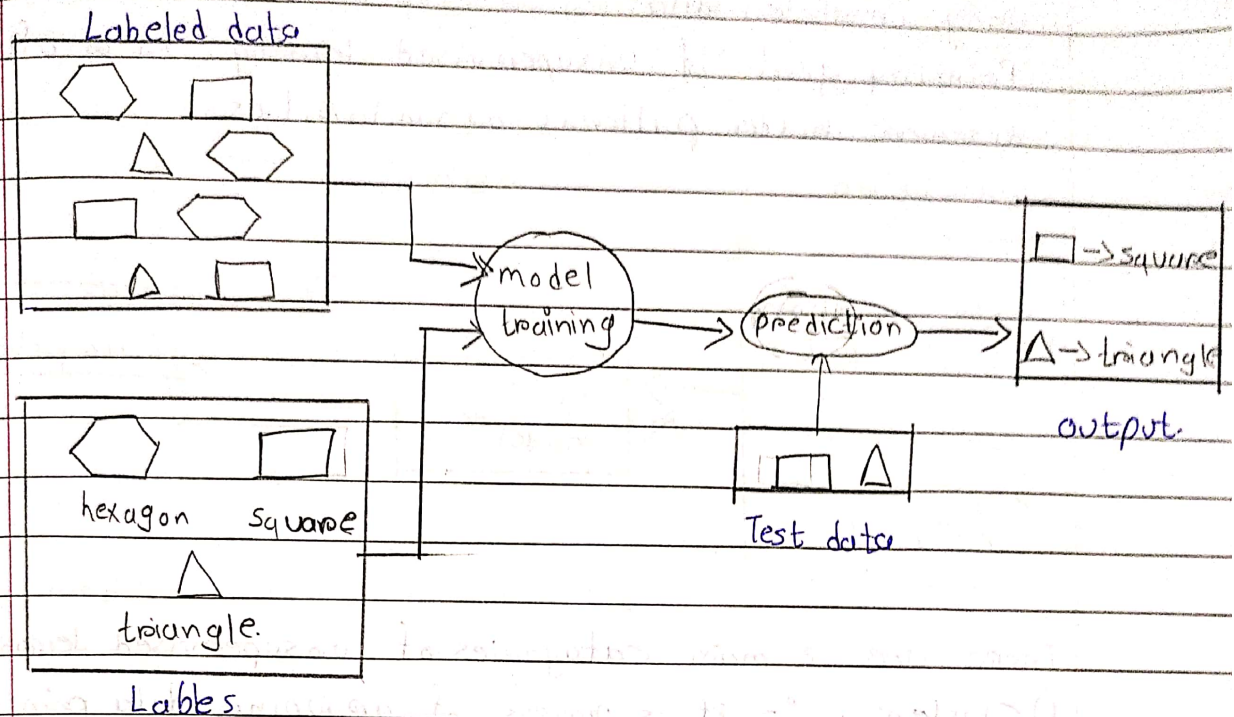


1) Supervised learning :-

- It is defined as when a model gets trained on a labelled dataset.
- The labelled data means some input data is already

tagged with correct output.

- Labeled datasets have both input and output.



There are main 2 categories of supervised learning.

1) Classification :- It predicts categorical target variables which represents discrete classes or labels. These algorithms map input to one of predefined classes.

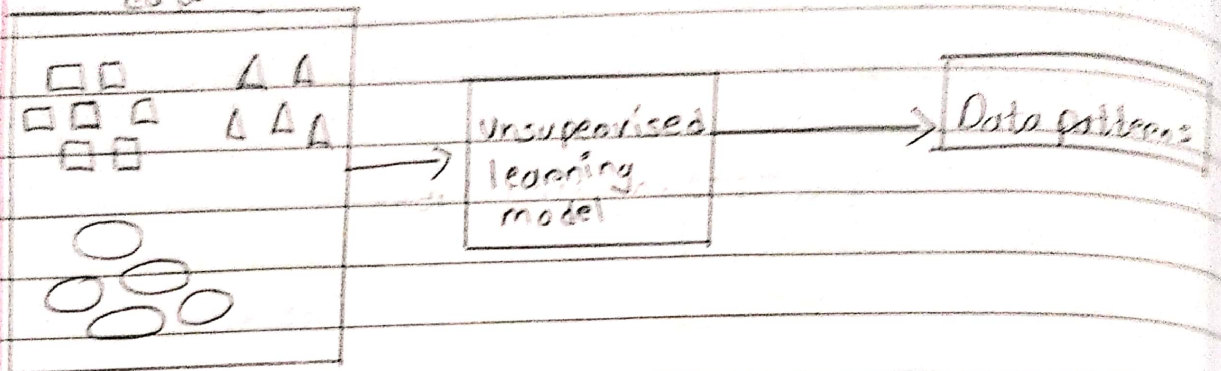
- Logistic Regression
- Support vector Machine
- Decision tree
- Naive Bayes

2) Regression :- It deals with predicting continuous target variables which represents numerical values.

- Linear Regression
- Polynomial Regression
- Ridge/Lasso Regression

② Unsupervised learning :-

- In this algorithm discovers patterns and relationships using unlabeled data.
- Primary goal of unsupervised learning is to often discover hidden patterns or similarities



There are 2 main categories of unsupervised learning:-

③ Clustering :- It is process of grouping data points into clusters based on their similarity.

- K-Mean clustering
- DBSCAN
- PCA

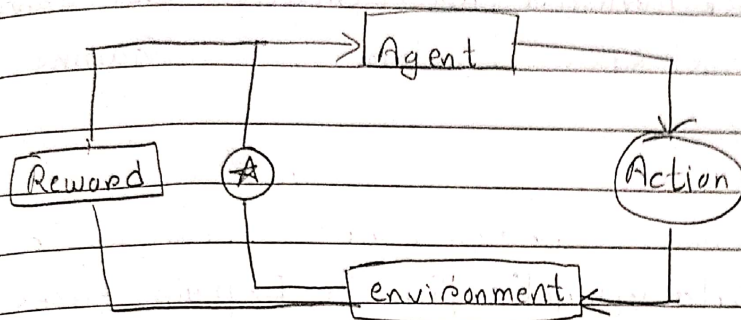
④ Association :- Association rule learning is a technique for discovering relationship between items and in dataset. It identifies rules that indicate the presence of one item implies the presence of another item with a specific probability.

③ Reinforcement learning

- Reinforcement learning is a feedback based Machine learning technique in which an agent learns to behave in an environment by performing the actions

and seeing the results of action.

- Trial, error and delay are the most relevant characteristics of reinforcement learning.



3) Compare Supervised, Unsupervised and Reinforcement ML.

Criteria	Supervised	Unsupervised	Reinforcement
Defination	Learn by labelled data.	Trained using unlabelled data	Works on interacting with environment.
Type of data.	Labelled	Unlabelled	No predefined data
Types of problem	Regression and classification	Association, clustering	Exploitation or Exploration
Supervision	Extra supervision.	No supervision.	No supervision
Aim	Calculate outcomes	Discovers patterns	Learn a series of action.

Algorithm	Logistic/Linear Regression, SVM, KNN, NB	K-Mean, PCA, DBSCAN	Q-Learning
Application	Risk Evaluation, Forecast Sales	Recommendation System.	Self driving Cars, Gaming

4) What is Data visualization ? Explain types of data for data visualization and types of Data Visualization approaches in details

⇒ Data visualization is the graphical representation of data to help identify patterns, trends and insights before applying machine learning models

Data visualization translates complex data set into visual formats that are easier for the human brain to comprehend

Types of data for DV :-

Numerical Data :-

1) Continuous data :- Data that can take infinite values
 Ex - temperature, weight.

2) Discrete Data :- Data that has distinct values
 Ex - no. of students, shoe size.

Categorical data :-

① Binary data :- Data with only two possible values
EX - Yes/No, Male/Female.

② Nominal data :- Data with categories that have no specific order.

EX - Colors, Countries

③ Ordinal data :- Data with categories that have meaningful Order.

EX - Ranking, levels

Types of Data visualization Approaches :-

Machine Learning may make use of wide variety of DV approaches, that include :-

① Line charts.

② Scatter plots

③ Bar charts

④ Heat Maps

⑤ Tree Maps

⑥ Box plots

Line charts :-

In line charts, each data point is represented by a point on the graph, and these points are connected by a line. We may find patterns and trends in the data across time by using line charts.

Scatter plots :-

A quick and efficient method of displaying the relationship between two variables is to use scatter plots. with one variable plotted on X-axis and the other variable on y-axis. each data point in a scatter plot is represented by a point on the graph.

Bar charts :-

Bar charts are common way of displaying categorical data. In a bar chart, each category is represented by a bar, with the height of the bar indicating the frequency of that category^{of} data.

Heat Maps :-

Heat Maps are type of graphical representation that displays data in a matrix format. The value of the data point that each matrix cell represents determines its hue. Heat maps are often used to visualize the correlation between variables.

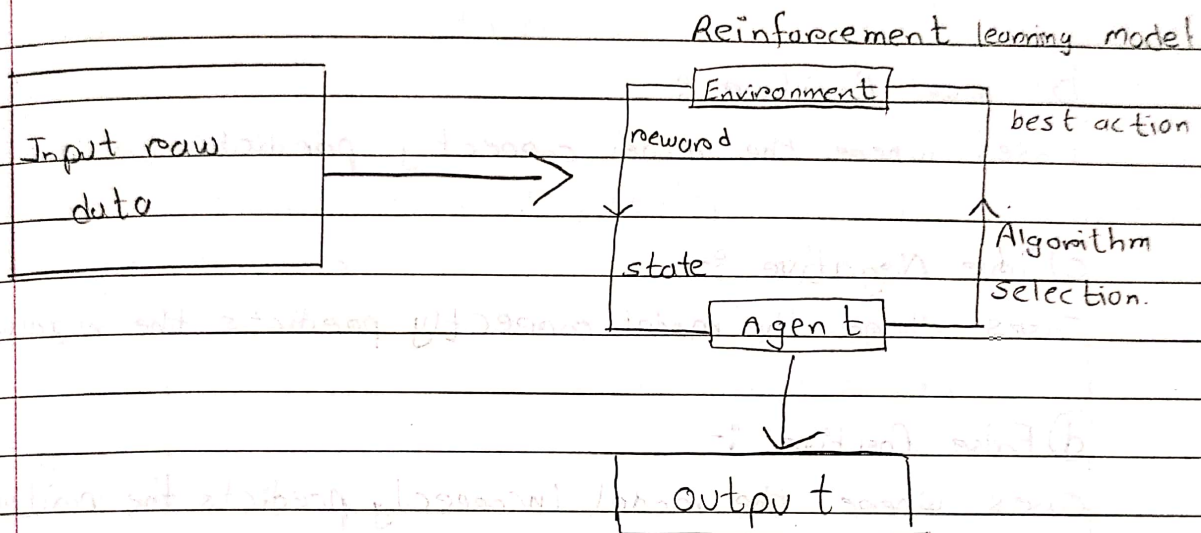
Box plots :-

Box plots are a graphical representation of the distribution of a set of data. In a box plot, the median is shown by a line inside the box, while the center box depicts the range of the data. The whiskers extend from the box to highest and lowest values in the data excluding outliers.

5) Explain Reinforcement Learning with suitable diagram.

⇒ Reinforcement learning is a feedback-based machine learning technique in which an agent learns to behave in an environment by performing the actions and seeing the results of actions.

For each good action, the agent gets positive feedback, and for each bad action, the agent gets negative feedback or penalty.



RL operates on the principle of learning optimal behavior through trial and error.

The agent takes actions within the environment, receives rewards or penalties and adjusts its behavior to maximize the cumulative reward.

⑤ Define the following terms :-

① Confusion matrix :-

A table used to evaluate the performance of a classification model by comparing predicted labels with actual labels. It consists of four values :-

True Positive (TP),

False Positive (FP),

True Negative (TN),

False Negatives (FN)

② True Positive :-

Cases where the model correctly predicts the positive case.

③ True Negative :-

Cases where the model correctly predicts the negative case.

④ False Positive :-

Cases where the model incorrectly predicts the positive class when the actual class is negative.

⑤ False Negative :-

Cases where the model incorrectly predicts the negative class when the actual class is positive.

⑥ Accuracy :-

The proportion of correct predictions out of the total number of cases

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

g) Precision :-

The proportion of correctly predicted positive cases out of all predicted positives

$$\text{precision} = \frac{TP}{TP + FP}$$

h) Recall :-

The proportion of actual positives correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}$$

i) F1-score :-

The harmonic mean of precision and recall, balancing both metrics.

$$F1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

j) Threshold :-

A decision boundary value used to classify predictions as positive or negative. Adjusting the threshold affects the trade-off between precision and recall.

k) False Positive Rate (FPR) :-

The proportion of actual negatives incorrectly classified as positive

$$FPR = \frac{FP}{TN + FP}$$

① AUC-ROC curve :-

A performance measurement that plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold values.

⑦ Calculate accuracy, precision, recall and F₁ score for given confusion matrix. Also identify TP, TN, FP and FN.

	Predicted No	Predicted Yes
Actual No	50	10
Actual Yes	5	100

$$TP = 100$$

$$TN = 50$$

$$FP = 10$$

$$FN = 5$$

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{150}{165} = 0.91$$

$$\text{precision} = \frac{TP}{TP + FP} = \frac{100}{110} = 0.91$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{100}{105} = 0.95$$

$$F_1 \text{ score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 0.93$$

8) I had taken random sample of 500 female, male customers, out of these 50 are actually female, our model predicted 100 total females. 45 of which are actually females.

a) Identify TP, TN, FP, FN and construct confusion matrix.

b) Calculate Accuracy, Precision, Recall and F1-score.

a) $TP = 45$

$TN = 5$

$FP = 55$

$FN = 395$

confusion matrix :-

	Predicted not female	Predicted female
Not female Actually male	395	55
Actually female	5	45

b) $Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{50}{500} = 0.1$

$Precision = \frac{TP}{TP + FP} = \frac{45}{100} = 0.45$

$Recall = \frac{TP}{TP + FN} = \frac{45}{440} = 0.10227$

$F1\ score = 2 \times \frac{Precision \times Recall}{Precision + Recall} = 0.167$

9) Apply Naive Bayes classification to classify whether fruit is "Apple" or "Orange", if the new fruit is "Red" Color.

Fruit ID	Color	Class
1	Red	Apple
2	Red	Apple
3	Green	Apple
4	Orange	Orange
5	Orange	Orange
6	Green	Apple

Step 1 :- Convert the data set into a frequency table.

Color	Apple	Orange	
Red	2	0	$P(\text{Red}) = \frac{2}{4}$
Green	2	0	$P(\text{Green}) = \frac{2}{4}$
Orange	0	2	$P(\text{Orange}) = \frac{2}{4}$

$$P(\text{Apple}) = \frac{4}{6} \quad P(\text{Orange}) = \frac{2}{6}$$

$$P(\text{Red} | \text{Apple}) = \frac{2}{4} = 0.5$$

$$P(\text{Red} | \text{Orange}) = \frac{0}{2} = 0$$

$$P(\text{Green} | \text{Apple}) = \frac{2}{4} = 0.5$$

$$P(\text{Green} | \text{Orange}) = \frac{0}{2} = 0$$

$$P(\text{Orange} | \text{Apple}) = \frac{0}{4} = 0$$

$$P(\text{Orange} | \text{Orange}) = \frac{2}{2} = 1$$

Probability of Apple when color is red.

$$\begin{aligned}P(\text{Apple}|\text{red}) &= \frac{P(\text{red}|\text{Apple}) * P(\text{Apple})}{P(\text{red})} \\&= \frac{0.5 * 0.66}{0.33} \\&= 1\end{aligned}$$

Probability of orange when color is red.

$$\begin{aligned}P(\text{Orange}|\text{red}) &= \frac{P(\text{Orange}|\text{red})}{P(\text{red})} \\&= \frac{P(\text{Red}|\text{orange}) * P(\text{orange})}{P(\text{red})} \\&= \frac{0 * 0.33}{0.33} \\&= 0\end{aligned}$$

$$P(\text{Apple}|\text{Red}) = 1$$

$$P(\text{Orange}|\text{Red}) = 0$$

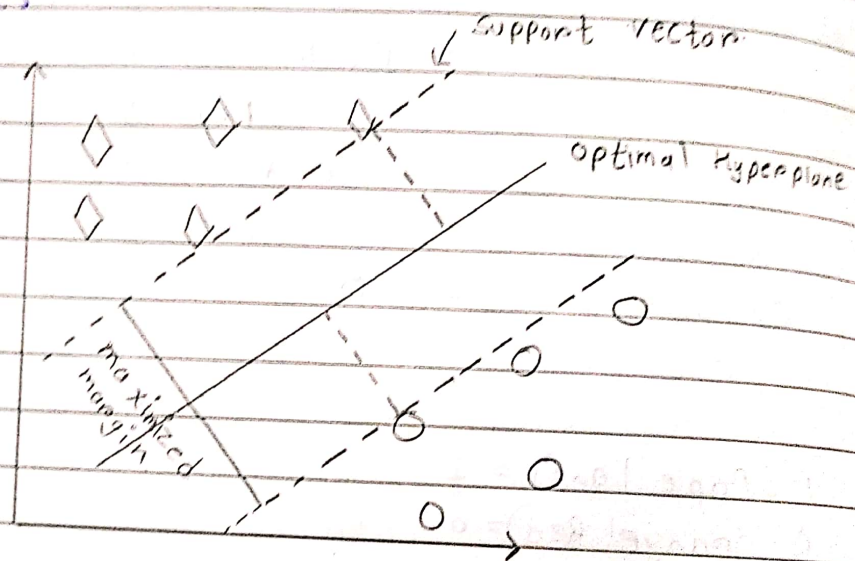
∴ new fruit is classified as Apple

Q2) Explain Support vector Machine with diagram and also discuss different types of kernel functions used in SVM.

→ SVM is a powerful supervised algorithm that works best on smaller datasets but on complex ones.

Support vector Machine abbreviated as SVM can be used for both regression and classification.

It is a supervised machine learning problem where we try to find a hyperplane that best separates the two classes.



SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors.

The distance between the vectors and hyperplane is called margin.

If there are 2 features then hyperplane will be a straight

Types of kernels used in SVM :-

1) Linear kernel :-

A linear kernel is the simplest form of kernel used in SVM. It is suitable when the data is linearly separable meaning that a straight line. It is used for text classification problems such as spam detection.

2) Polynomial kernel :-

The polynomial kernel allows SVM to model more complex relationships by introducing polynomial terms. It is useful when the data is not linearly separable but still follows a pattern. It is used in complex problems like image recognition where relationships between features can be non-linear.

3) Radial Basis function kernel (RBF) :-

The RBF kernel is the most widely used kernel in SVM. It maps the data into an infinite-dimensional space making it highly effective for complex classification problems.

4) Gaussian kernel :-

The Gaussian kernel is a special case of the RBF kernel and is widely used for non-linear data classification. It provides smooth and continuous transformations of data into higher dimensions. It is used when data has a smooth continuous distribution and requires a flexible boundary.

③ Sigmoid kernel

The sigmoid kernel is inspired by neural networks and behaves similarly to the activation function of a neuron. It is based on the hyperbolic tangent function and is suitable for neural networks and other non-linear classifiers. It is often used in neural networks and non-linear.

II] Write down the steps used in KNN classification process. How we select the value of K ?
⇒ The KNN algorithm compares a new data entry to the value in a given data set.

Based on its closeness or similarities in a given range (K) of neighbors, the algorithm assigns the new data to a class or category in the data set.

K-NN working steps:-

step 1 :- Assign a value to K

step 2 :- calculate the distance between the new data and all others existing data entries. Arrange them in ascending order.

step 3 :- Find the K nearest neighbors to the new entry based on calculated distances.

step 4 :- Assign the new data entry to the majority class in the nearest neighbors.

value of k :-

There is no particular way of choosing the value k , but here are some common conventions to keep in mind :-

- Choosing a very low value will most likely lead to inaccurate predictions
- The commonly used of k is 5.
- Always use an odd number as the value of k

Q2) Suppose to calculate the MAE, MSE and RMSE for given dataset.

X	Y	Pred_Y	Y - Y _{pred}	(Y - Y _{pred}) ²
10	150	150	0	0
20	250	290	-40	1600
30	350	340	10	100
40	450	490	-40	1600

$$MSE = \frac{1}{n} \sum (y - \bar{y})^2 = \frac{3300}{4} = 825$$

$$MAE = \frac{1}{n} \sum (y - \bar{y}) = \frac{190}{4} = 47.5$$

$$RMSE = \sqrt{MSE} = 28.7228$$

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Q3) What is Regression? Write any 10 applications of Regression. Explain types of regression models and equation for each in detail.

⇒ Regression is a statistical technique used in ML and data science to find relationship between a dependent and independent variable.

It helps in predicting outcomes, finding trends and understanding relationships between variables.

10 Applications of Regression :-

Stock market prediction

Sales Forecasting

Medical Diagnosis

House Price Prediction

Risk Management

Weather Forecasting

Customer churn Prediction

Traffic flow Analysis

Energy consumption Prediction

Employee performance Analysis

Types of regression models :-

1) Simple Linear regression :-

Models in a linear relationship between one dependent variable y and independent variable x .

$$y = mx + c$$

where m = slope of line

c = intercept

y = dependent variable

x = Independent variable

2) Multiple linear regression :-

Extends simple linear regression but here in the input variables we include some polynomial or higher degree terms. multiple independent variables

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

where y = dependent variable

x_i = independent variables

b_0 = intercept

b_i = Regression coefficients

3) Polynomial Regression :-

Extends linear regression by adding polynomial terms to capture non-linearity

$$y = b_0 + b_1x + b_2x^2 + \dots + b_nx^n$$

- 1A) Apply the simple linear Regression & find the MAE, MSE, RMSE for given dataset. Apply the Variance and Covariance method. Also calculate the value of y for $x = 125$

x	y	$\hat{y}$	$y - \hat{y}$	$(y - \hat{y})^2$	x^2
2	3	3.4	0.4	0.16	4
4	7	5.3	1.7	2.89	16
6	5	7.2	-2.2	4.84	36
8	10	9.1	0.9	0.81	64

$$MAE = \frac{1}{n} \sum_{i=1}^4 |y - \hat{y}|$$

$$= \frac{0.4 + 1.7 + 2.2 + 0.9}{4}$$

$$= 1.3$$

$$MSE = \frac{1}{n} \sum_{i=1}^4 (y - \hat{y})^2$$

$$= \frac{1}{4} (0.16 + 2.89 + 4.84 + 0.81)$$

$$= 2.175$$

$$RMSE = \sqrt{MSE} = \sqrt{2.175} \approx 1.475$$

$$\bar{x} = \frac{20}{4} = 5$$

$$\bar{y} = \frac{25}{4} = 6.25$$

$$\text{Var}(x) = \frac{1}{n} \sum (x_i - \bar{x})^2$$

$$= \frac{1}{n} [(2-5)^2 + (4-5)^2 + (6-5)^2 + (8-5)^2]$$

$$= \frac{20}{4}$$

$$= 5$$

$$\text{Cov}(x, y) = \frac{1}{n} [(2-5)(3-6.25) + (4-5)(7-6.25) + (6-5)(5-6.25) + (8-5)(10-6.25)]$$

$$= \frac{9.75 - 0.75 - 1.25 + 11.25}{4}$$

$$= \frac{19}{4}$$

$$= 4.75$$

$$a_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = \frac{4.75}{5} = 0.95$$

$$\begin{aligned} a_0 &= \bar{y} - a_1 \bar{x} \\ &= 6.25 - 0.95(5) \\ &= 6.25 - 4.75 \\ &= 1.5 \end{aligned}$$

$$y = a_0 + a_1 x$$

$$y = 0.95x + 1.5$$

$$\text{for } x = 125$$

$$y = (0.95)(125) + 1.5 = 120.25$$